

Facility: SALEM 1 & 2 Scenario No.: ESG-2 Op-Test No.: 11-01 NRC

Examiners: _____ Operators: _____

Initial Conditions: 75% power, MOL. Power was reduced 30 minutes ago due to a short duration oscillation on 21SGFP speed. PZR PORV 2PR1 was declared inoperable 3 hours ago due to control circuit problems, and its motor operated block valve 2PR6 is shut and power is removed from its breaker to comply with TSAS 3.4.5 action b.

Turnover: Maintain current power.

Event No.	Malf. No.	Event Type*	Event Description
1	RC0014B	I CRS RO	22 RC Loop Tavg Channel failure (TS)
2	SG0078C	C ALL	500 gpd SGTL on 23 SG. (TS)
3		R ALL	Power reduction
4	SG0078C	M ALL	SGTR on 23 SG
5	EL0134	C ALL	Loss of Off Site power during SGTR RCS cooldown (RO/CRS restart safeguards loads, PO re-establish RCS cooldown via SG atmospheric reliefs)
6	VL0298	C CRS RO	Only available PZR PORV 2PR2 fails to open results in SGTR without RCS pressure control
			CTs: 1. Isolate feedwater into and steam flow out of 23 SG, 2. Cooldown and maintain CET Target temperature.

* (N)ormal, (R)eactivity, (I)nstrument, (C)omponent, (M)ajor

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: SGTL / SGTR w no pressure control

SCENARIO NUMBER: 11-01 NRC ESG-2

EFFECTIVE DATE: See Below

EXPECTED DURATION: 80 minutes

REVISION NUMBER: 02

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

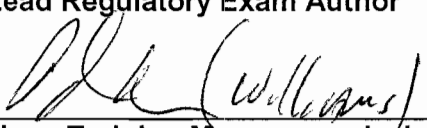
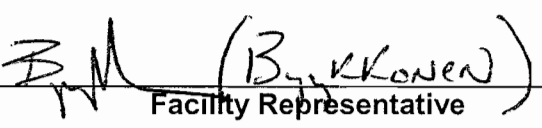
Revision Summary

New Issue for 2010 LORT Annual Exam

Rev. 1- Added TAC pp trip to beginning of scenario concurrent with SGTL.

Rev. 2 – 6/2012 Modified SGTL size from 90 to 500 gpd. Modified format to ILOT exam.

Removed TAC pump trip. Added Tavg channel failure. Modified temp at which loss of off-site power occurs (from 535 to 510) Removed 21 SGFP at idle, both SGFP's are in service

PREPARED BY:	<u>G Gauding</u> Lead Regulatory Exam Author	<u>08-04-2012</u> Date
APPROVED BY:	<u> (Williams)</u> Operations Training Manager or designee	<u>9/14/12</u> Date
APPROVED BY:	<u> (Bjorkonen)</u> Facility Representative	<u>9/5/12</u> Date

SCAN OF SIGNED SCENARIO COVER SHEET

I. OBJECTIVES

- A. Given a steam generator tube leak, take corrective action, IAW S2.OP-AB.SG-0001.
- B. Given the order or indications of a steam generator tube leak (SGTL), perform actions as the nuclear control operator to RESPOND to the tube leak in accordance with the approved station procedures.
- C. Given the order or indications of a steam generator tube leak (SGTL), DIRECT the response to the tube leak, in accordance with the approved station procedures.
- D. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.
- E. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures.
- F. Given the order or indications of a safety injection, perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- G. Given indication of a safety injection, DIRECT the response to the safety injection in accordance with the approved station procedures.
- H. Given the order or indications of a steam generator tube rupture (SGTR), perform actions as the nuclear control operator to RESPOND to the tube rupture in accordance with the approved station procedures.
- I. Given indication of a steam generator tube rupture (SGTR), DIRECT the response to the SGTR in accordance with the approved station procedures.
- J. Given the order or indications of a steam generator tube rupture (SGTR) without pressurizer pressure control, perform actions as the nuclear control operator to RESPOND to the SGTR in accordance with the approved station procedures.
- K. Given the indication of a steam generator tube rupture (SGTR) without pressurizer pressure control, DIRECT the response to the SGTR in accordance with the approved station procedures.
- L. During performance of emergency operating procedures, monitor the critical safety function status trees in accordance the EOP in effect.

II. MAJOR EVENTS

- A. 22 RCS Loop Th fails high
- B. 500 gpd SGTL
- C. SGTR
- D. Loss of Both SI pumps
- E. Loss of Off site power

III. SCENARIO SUMMARY

- A. The crew will take the watch at 75% power, middle of life. Power was reduced 30 minutes ago as a precaution due to a short duration speed oscillation on 21 SGFP. Both SGFPs remain in service. PZR PORV 2PR1 was declared inoperable 3 hours ago due to intermittent control circuit anomalies, and the PORV Block valve 2PR6 was shut and deenergized to comply with TSAS 3.4.5 action b.
- B. Shortly after the crew takes the watch, 22 loop Tavg channel fails high. The RO will place rod control in manual after determining stable Main Turbine load. The CRS will enter S2.OP-AB.ROD-0003, Continuous Rod Motion, remove the failed channel from service, and identify appropriate Tech Specs.
- C. A small (500 gpd) SGTL will ramp in on 23 SG. The CRS will enter S2.OP-AB.SG-0001, Steam Generator Tube Leak, and take actions to quantify the leak and minimize the spread of contamination in the secondary plant.
- D. The Crew identifies that the SGTL meets the criteria for Action Level 3 in S2.OP-AB.SG-0001, requiring the unit to be <50% power within one hour, and that Tech Spec 3.4.7.2 applies once the leak is >150 gpd.
- E. The CRS will direct a power reduction to be initiated. The RO will use boration and manual rod control while the PO reduces Main Turbine load.
- F. After the power reduction is initiated, the affected tube on 23 SG will rupture. Operators will identify the rising leak rate, and initiate a Rx trip and Safety Injection IAW CAS when conditions warrant.
- G. The crew will perform diagnostics in 2-EOP-TRIP-1, Reactor Trip or Safety Injection. 22 AFW pump pressure override protection fails and cannot be defeated. Operators will isolate AFW to 23 SG.
- H. The crew will transition to 2-EOP-SGTR-1, Steam Generator Tube Rupture. With the 23MS45 steam supply from 23 SG to 23 AFW pump still open, the CRS will secure 23 AFW pump if not performed previously, leaving 24 SG as the only generator receiving AFW flow. Once secured and with 23MS45 shut, the crew will reset 23MS52 and start 23 AFW pump.
- I. During the RCS cooldown in SGTR-1, a loss of off-site power will occur when the hottest CET reaches 510 degrees. The crew will perform the actions of Table C to restart safeguards loads, and re-establish the cooldown using the MS10s. 21 SI pump will not start if the crew attempts to start it, and 22 SI pump will trip if started.
- J. Once the target temperature has been reached, the crew will attempt to perform the RCS depressurization with the only available PORV since RCPs lost off site power and 2PR1 is unavailable. 2PR2 will not open. With no SI pumps running, the CRS will transition to SGTR-5.
- K. The scenario will terminate after the transition to SGTR-5 has been made.

IV. INITIAL CONDITIONS

___ IC- 242

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
___ 1	RH1 and RH2 C/T
___ 2	RH18s C/T
___ 3	VC1 and VC4 C/T
___ 4	RCPs (SELF CHECK)
___ 5	RTBs (SELF CHECK)
___ 6	MS167s (SELF CHECK)
___ 7	500 KV SWYD (SELF CHECK)
___ 8	SGFP Trip (SELF CHECK)
___ 9	23 CV PP (SELF CHECK)
___ 10	2PR6 C/T with power removed
___ 11	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
	2	EVENT ACTION: MONP254<10 // CONT BANK C GROUP POS COMMAND: PURPOSE: <update as needed>
	3	EVENT ACTION: monp187<510 // HOTTEST INCORE T/C TEMP.0 COMMAND: PURPOSE: <update as needed>
	4	EVENT ACTION: kbm06tcy // 23BF19 FW CONTROL VALVE-CLOSE COMMAND: DMF VL0448 PURPOSE: <update as needed>
	6	EVENT ACTION: kbm12tcy // 23BF13 FW INLET STOP VALVE CLOSE COMMAND: DMF VL0025 PURPOSE: <update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
01	SG0078C 23 STEAM GENERATOR TUBE RUPTURE	N/A	0	00:15:00	RT-1	0.5
02	EL0134 LOSS OF ALL 500KV OFF-SITE POWE	N/A	N/A	N/A	ET-3	
03	VL0298 2PR2 Fails to Position (0-100%)	N/A	N/A	N/A	RT-3	0.01
04	SJ0062B 22 SAFETY INJECTION PUMP TRIP	N/A	N/A	N/A	RT-5	
05	AF0182B 22 AFP PRESS OVRD PROT FAILS	N/A	N/A	N/A	N/A	
06	AN3735 AAS 735 FAILS - :21 TGA SUMP LEVEL HIGH	00:03:00	N/A	N/A	RT-2	AAS POINT FAILS/OVRD TO ON
07	AN3736 AAS 736 FAILS - :22 TGA SUMP LEVEL HIGH	00:03:10	N/A	N/A	RT-2	AAS POINT FAILS/OVRD TO ON
08	AN3737 AAS 737 FAILS - :23 TGA SUMP LEVEL HIGH	00:03:25	N/A	N/A	RT-2	AAS POINT FAILS/OVRD TO ON
09	AN3738 AAS 738 FAILS - :24 TGA SUMP LEVEL HIGH	00:03:37	N/A	N/A	RT-2	AAS POINT FAILS/OVRD TO ON
10	AN3739 AAS 739 FAILS - :25 TGA SUMP LEVEL HIGH	00:03:57	N/A	N/A	RT-2	AAS POINT FAILS/OVRD TO ON
11	VL0448 23BF19 Fails to Position (0-100%)	N/A	N/A	N/A	ET-2	56.62
12	VL0025 23BF13 Fails to Position (0-100%)	N/A	N/A	N/A	ET-2	100
13	RC0014B 22 Hot Leg RTD Avg Summator	N/A	N/A	N/A	RT-13	650

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
___ 01	PR34D PORV STOP VALVE 2PR6 TAGGED	N/A	N/A	N/A	N/A	TAGGED
___ 02	MS06A 23MS45 23 STM GEN STM SUP-23 AFP	N/A	N/A	N/A	RT-7	0
___ 03	AF01D 23 AUX FP TRIP RESET	00:10:00	N/A	N/A	RT-8	RESET

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
___ 01	AH01 F DI 21 SI PUMP STOP	00:00:05	N/A	N/A	ET-3	ON
___ 02	B511 A DI 22 AUX FEED PUMP-PRESS OVERRIDE DEFEAT	N/A	N/A	N/A	N/A	OFF

TAGGED EQUIPMENT:

SELF-CHECK	Description
___ 1.	RH1 and RH2 (C/T)
___ 2.	VC 1 and 4 (C/T)
___ 3.	RH 18's (C/T)
___ 4.	RCPs (SELF CHECK)
___ 5.	RT (SELF CHECK)
___ 6.	MS 167s (SELF CHECK)
___ 7.	500 KV SWYD (SELF CHECK)
___ 8.	2PR6 shut with power removed.
___ 9.	SGFP TRIP (SELF CHECK)

OTHER CONDITIONS:

Description
___ 1. None

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
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1. 22 RCS Loop Th Fails High

Simulator Operator: Insert RT-13 on direction from Lead Evaluator.

MALF: RC0014B, 22 Hot Leg RTD Avg Sum Fail

Final Value: 650

RO reports control rods inserting at 72 spm, verifies no runback in progress, gains concurrence from CRS and places control rods in manual.

CRS enters S2.OP-AB-ROD-3, Continuous Rod Motion.

RO reports rod motion stopped.

RO adjusts Rods in manual to control Tavg within 1.5° of program using Att. 1.

RO reports rod motion was inward.

RO reports no NI failures.

RO reports 22 loop Tavg failed high and no dilution in progress.

RO places Master Flow Controller to manual and adjusts charging flow to return PZR level

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	to program using Att. 2.		
	RO selects Deviation Defeat for Loop 22 ΔT and Tave.		
	RO selects recorders to channel other than 22 loop.		
	RO returns Master Flow Controller to auto when PZR level is on program.		
	RO reports control rods are above the RIL.		
	RO withdraws control rods to ARO or to establish Tave within 1.5°F of program.		
	RO verifies Tave within 1.5° program and restores the Rods to Auto.		
	CRS initiates removing 22 loop RCS temperature channel from service IAW SO.RPS-0002.		
	CRS enters TSAS(s) 3.3.1.1 Action 6, and 3.3.2.1 Action 19*.		
Proceed to next event after Tech Spec call is made at Lead Evaluators direction.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
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2. SGTL

Simulator Operator: Insert RT-1 on direction from Lead Evaluator.

**MALF: SG0078C 23 Steam Generator
Tube Rupture
Final Value: 0.5
Ramp: 15 minutes**

RO announces OHA A-6 RMS HI RAD OR TRBL as unexpected.

RO reports CRT shows 2R53C in alarm.

RO reports 2R53C reading and slowly rising.

CRS contacts Radiation Protection to perform SC.RP-TI.RM-0607(Q), Primary To Secondary Leak Rate Response IAW ARP.

CRS enters S2.OP-AB.SG-001 Steam Generator Tube Leak directly or enters S2.OP-AB.RAD-001, Abnormal Radiation.

Note: Go to downpower actions when Action Level 3 is identified.

CRS directs performance of S2.OP-AB.SG-001 CAS.

PO initiates trending of 2R19C and 2R53C.

Crew contacts Chemistry to:

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
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- Initiate sampling of 23 SG to confirm and quantify actual leak rate.
- Provide algorithm correlation factor (if required)
- Provide a rate-of-change for 23 SG primary to secondary leakage.

Role Play: When contacted, respond as Chemistry that a technician will be dispatched to the Control Room, and that sampling of 23 SG will be performed.

CRS dispatches an operator to deenergize TGA sumps.

Simulator Operator: Insert RT-2 when directed to deenergize TGA sumps. This RT includes a 3 minute delay before opening sump breakers. Report to control room when last of the 5 sump breakers has been deenergized.

MALFS:

AN3735 21 TGA Sump Level Hi

AN3736 22 TGA Sump Level Hi

AN3737 23 TGA Sump Level Hi

AN3738 24 TGA Sump Level Hi

AN3739 25 TGA Sump Level Hi

Delays: 3:00, 3:10, 3:25, 3:37, 3:57

Note: PZR level may be lowering if rods were not resored to ARO. (Programmed level lower due to lower Tavg.)

RO reports PZR level is stable. **IF** RO reports PZR level is lowering, RO will swap to a

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	centrifugal charging pump.		
	RO reports unit is in Mode 1.		
	Crew identifies 2R53C, 2R19C and 2R15 rising.		
	PO reports affected SG is 23.		
	PO raises 23MS10 setpoint to 1045 psig.		
	PO closes or checks closed 23GB4, 23MS7, and 23MS18.		
	CRS dispatches an operator to shut 23MS45, and enters TSAS 3.7.1.3 for 23 AFW pump.		
Simulator Operator: Do NOT shut 23MS45 until directed later in scenario.			
	CRS dispatches an operator to re-align SGBD and MS sampling to Waste System.		
CAS Action Level 3 required actions start here.			
	CRS determines that the CAS indicates that Action Level 3 is present IAW Step 6.2.A based on:		
	- Leak rate is ≥ 75 gpd AND - The rate of change of the leakrate is ≥ 30 gpd/hr.		
	CRS determines a power reduction to ≤ 50 % must be performed within 1 hour, and a		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	shutdown to Mode 3 (Hot Standby) within the next 2 hours.		
	CRS enters TSAS 3.4.7.2.c		
Note: If time does not permit Tech Spec identification, post scenario follow up question will be required.			
	RO calculates boron addition required for power reduction to 50%.		
	CRS orders a power reduction at a rate which will ensure power is <50% within one hour.		
	CRS enters S2.OP-AB.LOAD-0001, Rapid Load Reduction to perform the power reduction.		
Simulator Operator: MODIFY MALF SG0078C to 350 with a RAMP of 5 minutes after power reduction has commenced at direction of Lead evaluator.			
	RO reports reflash of OHA A-6, and CRT indicates 2R15 in alarm.		
	RO reports indications of worsening tube leak on 23 SG.		
OHA A-6 will reflash several more times as 2R15, 2R19C, and 2R41D alarm.			
	CRS determines CAS actions for rising SG NR		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
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level IAW CAS 1.0 are true.

CRS directs the RO to trip the Rx, confirm the trip, and initiate a Safety Injection.

Simulator Operator: Ensure **ET-2** is TRUE upon the Rx trip. This fails the 23BF19 to 56% open and 23BF13 full open.

RO trips the Rx, confirms the trip, and initiates a Safety Injection.

Evaluators log time of Rx Trip here:

____:____:____

RO performs immediate actions of TRIP-1:

Verifies Rx tripped.
Verifies Rx trip confirmation
Trips the Main Turbine.
Verifies off site power available to vital busses
Verifies SI initiation.

CRS reads immediate action steps to RO who confirms their performance.

Crew commences monitoring TRIP-1 CAS.

RO reports SEC loading is not complete for all vital busses, but all available equipment started.

PO reports all AFW pumps are running, but 22

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
AFW flow control is complicated by the fact that a SGTR is present on 23 SG, but the 23MS45 has not been shut yet. Additionally, 22 AFW pump is not supplying flow to 21 and 22 SGs because its Pressure Override circuit has failed. Shortly after the Rx is tripped, SG level will recover in at least one SG so that AFW flow can be lowered < 22E4. TRIP-1 does not direct tripping 23 AFW pump unless 2 MDAFW pps are running, which in this case is not true, since a running pump would be supplying flow. Detailed observation of the crew during AFW flow decisions should be made by the evaluators.	AFW pump is not supplying flow even though its discharge pressure is high enough. PO requests, and receives, permission to depress Pressure Override Defeat for 22 AFW pump, which has no effect on AFW flow. PO reports 23 AFW pump is running, and that an unmonitored release is occurring from the TDAFW pump steam discharge with 23MS45 not shut yet.		
CT# 1: (E-3-B) Isolate AFW to the ruptured SG within 10 minutes of entry into TRIP-1 and subsequently close 23MS167, 23MS18, 23MS7 and 23GB4 in SGTR-1.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
SAT UNSAT			

PO requests to throttle AFW flow, and isolates AFW flow to 23 SG by closing 23AF11 and 23AF21.

CRS directs PO to maintain total AFW flow >22E4 lbm/hr while throttling AFW flow.

RO reports 23BF19 and 23BF13 are NOT shut when performing Safeguards Valve Alignment.

CRS directs PO to shut 23BF19 and 23BF13 from the control console.

Simulator Operator: Ensure **ET-4** is TRUE when the 23BF19 Close PB is depressed. This will remove the malfunction keeping 23BF19 open.

Simulator Operator: Ensure **ET-6** is TRUE when the 23BF13 Close PB is depressed. This will delete the malfunction that is keeping the 23BF13 open.

PO shuts 23BF19 and 23BF13.

RO reports normal containment pressure.

CRS determines no MSLI is required.

CRS directs SM to refer to the ECG.

PO reports all 4KV vital busses energized from

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	off site power.		
	RO reports control room ventilation in accident pressurized mode.		
	RO reports 2 CCW pumps are running.		
	RO reports ECCS is injecting as expected for current RCS pressure.		
	PO reports AFW flow and SG NR level status.		
	RO reports all RCPs are in operation.		
	RO reports MSLI is not required.		
	RO reports RTBs are open, PORVs shut, PORV block valve 2PR7 open, 2PR6 shut and C/T, and PZR spray valves operating as expected for current RCS pressure.		
	RO reports RCS pressure > 1350 psig.		
	RO maintains seal injection flow to all RCPs.		
	PO reports no faulted SGs.		
	PO reports 23 SG is ruptured based on NR level.		
	CRS transitions to SGTR-1.		
	Crew commences monitoring of CFSTs.		
	RO maintains seal injection flow to all RCPs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports 23 SG is ruptured, and 23MS10 is set at 1045 psig.		
	PO reports 23MS10 operating as expected for current pressure.		
	PO shuts 23MS167 and ensures 23MS7, 23MS18, and 23GB4 are shut.		
	PO reports 23 SG is ruptured.		
	PO reports 23 AFW pump is NOT only source of AFW.		
	PO lowers 23 AFW pump speed to minimum and trips 23 AFW pump.		
	PO stops 23 AFW pump.		
Simulator Operator: When 23 AFW pump is stopped, and an operator has previously been dispatched, insert RT-7 to close 23MS45 and call control room to inform them it is shut. REMOTE: MS06A 23MS45 23 STM GEN STM SUP-23 AFP			
	CRS sends an operator to reset 23MS52 when all SG NR levels are > 15%.		
Simulator Operator: Insert RT-8 to reset 2MS52. RT includes a 10 minute delay prior to resetting MS52.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
REMOTE: AF01D 23 AUX FP TRIP RESET Final Value: RESET	CRS checks on status of MS sampling valves which were directed to be shut in AB.SG, or if not previously directed to be closed, directs closure of 2SS333. If previously dispatched to shut 2SS333, then when contacted by crew report 2SS333 is shut. PO reports 23 SG is isolated from intact SGs, 23 SG NR level is >9%, and feed flow is isolated to 23 SG. RO reports power is C/T to 2PR6, and CRS orders power restored. Note: Do NOT restore power to 2PR6 during scenario. If asked later in scenario, state the breaker appears to be mechanically bound in the open position. RO reports 2PR2 is operating correctly in response to PZR pressure. PO reports no faulted SGs. RO resets SI and Phase A isolations, and opens CA330's. PO resets all SECs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO stops both RHR pumps. PO reports 23 SG is isolated and >375 psig. CRS dispatches an operator to shift Gland Sealing steam to Unit 1. CRS determines target temperature for RCS cooldown is 503 degrees. PO commences RCS cooldown by using Steam Dumps in MS Pressure Control-Manual at 25% demand. PO Bypasses Tavg on steam dump control when Tavg reaches 543°F.		
Simulator Operator: Ensure ET-3 is TRUE when the hottest CET reaches 510 degrees. This inserts the loss of off-site power. MALF: EL0134 Loss of All Offsite 500KV Power I/O AH01 OVDI 21 SI Pump_STOP			
	CRS recognized loss of off-site power and returns to Step 10 CAS to perform actions of Table C. PO verifies all available equipment started for each SEC. PO resets all SECs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
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CRS directs starting of Safeguards loads.

Simulator Operator: When CRS directs start of SI pumps, insert **RT-5** after 22 SI pump has been started.

MALF: SJ0062B 22 Safety Injection Pump trip.

Note: 21 SI pump will not start.

CRS directs the cooldown to be re-established using intact SGs 21, 22 and 24MS10's fully open.

RO shuts charging pump mini flows when RCS pressure lowers to 1500 psig.

CT #2 (E-3-B): Establish/ maintain RCS temperature to maintain minimum required subcooling.

SAT _____ UNSAT _____

PO stabilizes hottest CET temperature less than 503 degrees.

PO reports 23 SG pressure is stable or rising or at least 250 psig above intact SG pressure.

RO reports adequate subcooling.

RO reports RCPs are stopped and normal

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
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PZR spray is not available.

Simulator Operator: Insert **RT-3** when Step 18 PORV status is read PRIOR to crew attempting to open 2PR2.

MALF: VL0298 2PR2 fails to position (0-100%)

Final Value: 0.01

RO reports 2PR2 PZR PORV is available.

Note: If asked about restoring power to 2PR6, report that there has been no progress in determining what is keeping the breaker from being closed.

RO reports 2PR2 will not open.

CRS returns to step 18 and answers NO to if a PORV is available.

Simulator Operator: IF CRS did not start SI pumps subsequent to the Blackout, and starts them here, THEN insert **RT-5** after 22 SI pump has been started.

MALF: SJ0062B 22 Safety Injection Pump trip.

Note: 21 SI pump will not start.

RO reports no SI pumps are running.

RO reports no SI pumps running if start attempt is made.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	CRS transitions to SGTR-5.		
	PO reports 23 SG NR level.		
	RO reports no RCPs are running due to loss of off-site power.		
	RO reports no PZR PORV available since 2PR6 is shut and deenergized to comply with TS 3.4.5.b		
	RO reports neither SI pump will run.		
	PO maintains AFW flow 9-33% with any SG NR level >9%.		
	RO reports PZR level >11% and RCS subcooling >0°.		
	PO reports >22E4 lbm/hr AFW flow is available.		
	RO reports RVLIS level >57% Full Range.		
	PO reports 23 SG ruptured.		
	RO stops all but 21 OR 22 charging pumps.		
Terminate scenario after reduction to single charging pump has been performed at Lead Evaluators direction.			

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-AB.RAD-0001, Abnormal Radiation
- F. S2.OP-AB.SG-0001, Steam Generator Tube Leak
- G. 2-EOP-TRIP-1, Rx Trip or Safety Injection
- H. 2-EOP-CFST-1, Critical Safety Function Status Trees
- I. 2-EOP-SGTR-1, Steam Generator Tube Rupture
- J. 2-EOP-SGTR-5, Steam Generator Tube Rupture w/o Pressure Control

ATTACHMENT 1 UNIT TWO PLANT STATUS TODAY

MODE: 1 POWER: 75 RCS BORON: 915 MWe 890

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Core Burnup is 5000 EFPD

Control Bank D rods are in auto at 154 steps

Power was reduced 30 minutes ago as a precaution due to a short duration speed oscillation on 21 SGFP

Xenon building in at 80 pcm / hr

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

3.4.5. Action b- 2PR1 expires 69 hours from now

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

None

ABNORMAL PLANT CONFIGURATIONS:

2PR6 shut and power C/T.

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

No penalty minutes in the last 24 hrs.

PRIMARY:

PZR PORV 2PR1 was declared inoperable 3 hours ago due to intermittent control circuit anomalies.

SECONDARY:

21 SGFP speed problem has not been investigated.

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

None

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
with Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.

ATTACHMENT 3

CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4 SIMULATOR SCENARIO REVIEW CHECKLIST

Note: Attach a separate copy of this form to each scenario reviewed. This form is used as guidance for the examination team as they conduct their review for the proposed scenarios.

SCENARIO IDENTIFIER: 11-01 NRC ESG-2 **REVIEWER:** E Gallagher

Initials	Qualitative Attributes
EG	1. The scenario has clearly stated objectives in the scenario.
EG	2. The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
EG	3. The scenario consists mostly of related events.
EG	4. Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
EG	5. No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
EG	6. The events are valid with regard to physics and thermodynamics.
EG	7. Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
EG	8. The simulator modeling is not altered.
EG	9. All crew competencies can be evaluated.
EG	10. The scenario has been validated.
EG	11. If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
EG	12. ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4(cont) SIMULATOR SCENARIO REVIEW CHECKLIST

Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

Initial		Quantitative Attributes
GG	7	Total malfunctions inserted: 5-8
GG	3	Malfunctions that occur after EOP entry: 1-2
GG	2	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	1	EOP contingencies requiring substantive actions: 0-2
GG	2	Crew Critical Tasks: 2-3
Comments:		

Attachment 5
ESG Critical Tasks

11-01 NRC ESG-2

CT #1: (E-3-A) Isolate AFW to the ruptured SG within 10 minutes of entry into TRIP-1 and subsequently close 23MS167, 23MS18, 23MS7 and 23GB4 in SGTR-1.

BASIS: Failure to isolate the ruptured SG causes a loss of differential pressure between the ruptured SG and the intact SGs. Upon a loss of differential pressure, the crew must transition to a contingency procedure that constitutes an incorrect performance that "...necessitates the crew taking compensating actions which complicates the event mitigation strategy..."

CT #2: (E-3-B) Establish/Maintain RCS temperature so a transition from SGTR-1 does not occur because Tave is too high to maintain minimum subcooling, OR below the RCS temperature that causes an extreme challenge (Red) or severe (Purple) challenge to the subcriticality and/or the integrity CSF.

BASIS: Failure to establish and maintain the correct RCS temperature during a SGTR leads to a transition from E-3 to a contingency procedure, which constitutes an incorrect performance that "...necessitates the crew taking compensating actions which complicates the event mitigation strategy..."

ATTACHMENT 6**ESG – PSA RELATIONSHIP EVALUATION****11-01 NRC ESG-2****SALEM ESG - PRA RELATIONSHIPS EVALUATION FORM****EVENTS LEADING TO CORE DAMAGE**

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
N	TRANSIENTS with PCS Unavailable	N	Loss of Service Water
Y	Steam Generator Tube Rupture	N	Loss of CCW
Y	Loss of Offsite Power	N	Loss of Control Air
N	Loss of Switchgear and Pen Area Ventilation	N	Station Black Out
N	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
N	Containment Sump Strainers	N	Gas Turbine
N	SSWS Valves to Turbine Generator Area	N	Any Diesel Generator
N	RHR Suction Line valves from Hot Leg	Y	Auxiliary Feed Pump
N	CVCS Letdown line Control and Isolation Valves	N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
Y	Cooldown the RCS and depressurize the system
Y	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed